Journal Pre-proof

Ferroptosis drives the sex-specific gonadal toxicity of microcystin-LR in mussels Hyriopsis cumingii with doubly uniparental inheritance

Liusiqiao Tang, Xuechun Wang, Yang Gu, Jianwei Wang, Yingrui Mao, Qinyun Hu, Guiling Wang

PII: S2590-1826(26)00104-9

DOI: https://doi.org/10.1016/j.enceco.2026.05.014

Reference: ENCECO 552

To appear in: *Environmental Chemistry and Ecotoxicology*

Received date: 27 November 2025

Revised date: 15 May 2026

Accepted date: 19 May 2026

Please cite this article as: L. Tang, X. Wang, Y. Gu, et al., Ferroptosis drives the sex-specific gonadal toxicity of microcystin-LR in mussels Hyriopsis cumingii with doubly uniparental inheritance, *Environmental Chemistry and Ecotoxicology* (2024), https://doi.org/10.1016/j.enceco.2026.05.014

Ferroptosis Drives the Sex-Specific Gonadal Toxicity of Microcystin-LR in Mussels *Hyriopsis cumingii* with Doubly Uniparental Inheritance

Liusiqiao Tang [a, b, c, 1], Xuechun Wang[a, b, c, 1], Yang Gu[a, b, c], Jianwei Wang[a, b, c], Yingrui Mao[a, b, c] , Qinyun Hu[a, b, c], Guiling Wang[a, b, c, *].

[a] Key Laboratory of Freshwater Aquatic Genetic Resources, Ministry of Agriculture and Rural Affairs, Shanghai Ocean University, Shanghai 201306, China

[b] National Demonstration Center for Experimental Fisheries Science Education, Shanghai Ocean University, Shanghai 201306, China

[c] Shanghai Engineering Research Center of Aquaculture, Shanghai 201306, China

* Corresponding author: Guiling Wang

E-mail address: yhwuwang2008@126.com (G. Wang).

[1]These authors contributed equally to this work.

Abstract: The reproductive toxicity of microcystin-LR (MC-LR) threatens declining freshwater mussels, many of which have a doubly uniparental inheritance (DUI) system that may confer sex-specific vulnerability. However, whether this toxicity is driven by ferroptosis and mediated by the DUI system remains unknown. Therefore, using the mussel *Hyriopsis cumingii*, we investigated whether ferroptosis mediates MC-LR-induced gonadal toxicity and sex-specific responses under DUI. Male and female mussels were exposed to environmentally relevant concentrations of MC-LR (0, 10, 40, and 70 μg/L) for up to 4 days. Gonadal tissues were subsequently analyzed for oxidative stress markers, mitochondrial function, ferroptosis-related gene expression, and ultrastructural changes. MC-LR exposure induced a classic time- and concentration-dependent oxidative stress response, characterized by the collapse of antioxidant defenses (SOD, CAT, GSH) and the accumulation of ROS and MDA. This was accompanied by mitochondrial dysfunction, evidenced by a sharp decline in mitochondrial membrane potential ($\Delta\Psi_m$) and ultrastructural damage. Intracellular Fe^{2+} levels increased, the expression of the key anti-ferroptotic gene *GPX4* was suppressed, while pro-ferroptotic genes *ACSL4* and *PTGS2* were significantly upregulated. Principal component analysis revealed distinct sex-specific clustering, demonstrating that male gonads were strongly associated with markers of oxidative damage and ferroptosis, whereas female gonads maintained a stronger antioxidant capacity. These results demonstrate that ferroptosis underlies MC-LR reproductive toxicity, with sex-specific vulnerability linked to the DUI system, providing insights for ecological risk assessment and conservation of freshwater bivalves.

Keywords: *Hyriopsis cumingii*; Microcystin-LR; Oxidative stress; Ferroptosis; Doubly uniparental inheritance; Gonad.

1. Introduction

Freshwater bivalves are a key functional group maintaining the structure and function of global freshwater ecosystems. Through efficient filter-feeding, they contribute to water purification, nutrient cycling, and energy flow, making their populations vital indicators of ecosystem health [1] . However, freshwater bivalves face a severe global decline due to multiple stressors, including habitat destruction and pollution. Unionid mussels are among the most imperiled taxa globally, with over 143 species listed as extinct, critically endangered, endangered, or vulnerable[2-4] . Although these stressors are known, the intrinsic biological mechanisms that translate external pressures into population collapse remain poorly understood.

While mitochondrial transmission in the majority of metazoans follows strict maternal inheritance (SMI), in which paternal mitochondria are actively eliminated after fertilization to maintain mitochondrial homoplasmy [5], a subset of bivalve species exhibits a notable exception known as doubly uniparental inheritance (DUI). This system is predominantly documented in the order Unionida and is also widely reported in representatives of Mytilida and Venerida [6]. This system maintains two distinct and heritable mitochondrial lineages: the F-type (female-transmitted) and M-type (male-transmitted). F-type mtDNA is passed to all offspring, whereas M-type mtDNA is transmitted via sperm exclusively to male progeny [7]. Consequently, females are homoplasmic (F-type only), while males are heteroplasmic, with their gonads predominantly harboring M-type mitochondria [8]. M-type genomes exhibit faster evolutionary rates, which may lead to reduced efficiency of the electron transport chain and higher endogenous reactive oxygen species (ROS) leakage. This suggests that male gonads may exist at a higher basal state of oxidative stress and be more susceptible to exogenous pollutants that induce oxidative damage [9, 10]. Therefore, this inherent, sex-specific difference dictated by DUI represents a critical but overlooked factor in their vulnerability.

Cyanobacterial blooms, a global threat driven by eutrophication, release microcystins (MCs), a family of structurally related cyclic heptapeptide hepatotoxins with more than 250 structural variants identified to date [11]. Among them, microcystin-LR (MC-LR) is widely used as a representative model toxin in mechanistic studies because of its high prevalence. MC-LR concentrations have been reported to reach approximately 70 μg/L in eutrophic freshwater systems [12] . Extensive studies have demonstrated that MC-LR not only can directly induce acute toxicity and even mortality in animals, but also can cause damage to multiple physiological systems through various mechanisms, including inhibition of protein phosphatase activity, induction of oxidative stress, and disruption of cellular signaling pathways. These adverse effects involve several vital organs and systems, such as the liver, kidney, nervous system, and reproductive system [13–16]. Notably, reproductive toxicity plays a critical role in the long-term ecological risk assessment of environmental contaminants, as it represents an important biological link between individual-level impairment and population-level decline. Previous studies have shown that the gonad is one of the key target organs of MC-LR, and its exposure can induce pronounced reproductive toxicity in a variety of animal models, including mice [17], *Danio rerio* [18], *Pelophylax nigromaculatus* [19], and *Macrobrachium rosenbergii* [20]. These toxic effects are mainly manifested as gonadal histopathological damage, disruption of sex hormone homeostasis, impaired gametogenesis, and reduced reproductive success.

Recently, ferroptosis, an iron-dependent form of regulated cell death characterized by lethal lipid peroxidation, has been identified as a key mechanism in oxidative stress-induced tissue damage [21]. Its hallmarks include glutathione (GSH) depletion and inactivation of glutathione peroxidase 4 (GPX4), leading to the fatal accumulation of lipid peroxides catalyzed by Fe^{2+}. Emerging evidence indicates ferroptosis is involved in the toxicological responses of aquatic animals to pollutants, including MC-LR [22]; Notably, a proteomic study on mammals has demonstrated that MC-LR induces gastric impairment associated with ferroptosis [23], supporting the broader relevance of this cell death pathway in MC-LR toxicity. As hubs for ROS production and iron metabolism, mitochondria play a central role in regulating ferroptosis [24]. We, therefore, hypothesized that ferroptosis is a primary mechanism underlying MC-LR-induced gonadal injury in bivalves.

The freshwater pearl mussel *Hyriopsis cumingii*, which follows a DUI system, was selected as the model organism. The primary aim of this study was to investigate the sex-specific gonadal toxicity of MC-LR in the mussel. Male and female mussels were exposed to environmentally relevant concentrations of MC-LR (10, 40, and 70 μg/L) to assess their toxicological responses. After exposure, a suite of indicators was measured at biochemical and molecular levels, including markers for oxidative stress, mitochondrial function, and ferroptosis. It was hypothesized that due to the unique Doubly Uniparental Inheritance (DUI) system, male gonads would exhibit heightened susceptibility to MC-LR-induced ferroptosis compared to female gonads. This study is the first attempt to mechanistically link the DUI genetic framework to a specific cell death pathway (ferroptosis) as a driver of sex-specific vulnerability in bivalves. These findings provide a novel toxicological basis for understanding the decline of freshwater mussel populations and support the ecological health risk assessment of cyanotoxins.

2. Materials and methods

2.1 Experimental materials

All animal procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of Shanghai Ocean University, China (Approval No. SHOU-DW-2025-007). Twelve-month-old adult mussels were obtained from Zhejiang Aquaculture Farm and acclimated to laboratory conditions for one week. Following sex determination via microscopic examination of gonad smears, 306 healthy individuals (shell length: 13.0 ± 1.1 cm; body weight: 80.41 ± 14.27 g) were selected for the experiment.

2.2 Experimental design

Microcystin-LR (MC-LR; analytical standard, purity ≥ 95%; Yuanye Bio-Technology Co., Ltd, Shanghai) was dissolved in methanol to a stock concentration of

1 mg/mL. For the exposure, mussels were maintained in 10 L tanks containing MC-LR at final concentrations of 10, 40, or 70 μg/L. The methanol vehicle volume was equalized to 0.007% (v/v) across all groups, including a solvent-only control (0 μg/L MC-LR).

In addition to a pre-exposure control (0 d), gonad tissues (n=9 biological replicates per group) were sampled at 1, 2, 3, and 4 d. Collected tissues were either flash-frozen and stored at –80 °C or preserved in 2.5% glutaraldehyde in phosphate buffer (0.1 M, pH 7.4). Exposure media were renewed every 24 h.

2.3 Quantification of Microcystin-LR

MC-LR concentrations in gonad tissues were determined using a commercial ELISA kit (Yuanju, China) according to the manufacturer's protocol. The sensitivity (limit of detection, LOD) of the kit was 1.0 ng/mL, and the limit of quantification (LOQ), based on the lowest concentration standard, was 1.0 ng/mL. Tissue samples (~100 mg) were homogenized in 500 μL of ice-cold 1× PBS and centrifuged (12,000 × g, 15 min, 4 °C). The supernatant was analyzed in triplicate, and absorbance was read at 450 nm on a microplate reader (BioTek Synergy 2, USA). Concentrations were calculated from a standard curve. Results were normalized to the total protein content of the supernatant, as measured by a BCA assay, and are expressed as ng/mg protein.

2.4 Measurement of Physiological and Biochemical Indicators

The activities of superoxide dismutase (SOD) and catalase (CAT), along with the levels of reactive ROS, malondialdehyde (MDA), reduced GSH, ferrous iron (Fe^{2+}), and mitochondrial membrane potential ($\Delta\Psi_m$) were quantified using commercial kits (Solarbio, Beijing, China). All absorbance and fluorescence measurements were performed on a BioTek Synergy 2 microplate reader (Winooski, VT, USA). For SOD activity, 20 μL of sample was incubated with a working solution containing WST-1 and xanthine oxidase for 20 min at 37 °C, and absorbance was measured at 450 nm. CAT activity was determined by monitoring the change in absorbance at 240 nm for 1 min following the addition of 10 μL of sample to a solution of 0.02 M H_2O_2 in 0.05 M phosphate buffer (pH 7.0). GSH content was quantified using the DTNB method, with absorbance measured at 412 nm. ROS levels were determined via the oxidation of 2',7'-dichlorofluorescein diacetate (DCFH-DA), with fluorescence measured at 488 nm excitation and 525 nm emission. MDA content was assayed using the thiobarbituric acid (TBA) method by reading the absorbance of the MDA-TBA adduct at 532 nm. Fe^{2+} levels were determined based on the formation of the Fe^{2+}-ferrozine complex, with absorbance read at 593 nm. The $\Delta\Psi_m$ was assessed with the JC-1 dye by recording the ratio of red (aggregates) to green (monomers) fluorescence. All results were normalized to total protein concentration, which was determined using the bicinchoninic acid (BCA) assay.

2.5 Quantitative Real-Time PCR (qRT-PCR) Analysis

Total RNA was isolated from samples using TRIzol™ Reagent (Thermo Fisher, **America**). The extracted RNA was then reverse-transcribed into cDNA using the PrimeScript™ RT Reagent Kit with gDNA Eraser (Takara, Japan). Quantitative real-time PCR was performed on a CFX96 Real-Time PCR Detection System (Bio-Rad, America) using TB Green® Premix Ex Taq™ II (Tli RNaseH Plus) (Takara Bio). The expression of target genes was normalized to the reference gene *EF1α*. All primer sequences used in this study are listed in **Table S1**.

2.6 Transmission Electron Microscopy (TEM)

Immediately after dissection, gonadal tissues were fixed in 2.5% glutaraldehyde in 0.1 M phosphate buffer (pH 7.4) for 48 h at 4℃. After rinsing, samples were post-fixed in 1% osmium tetroxide (OsO_4) for 2 h. The tissues were then dehydrated through a graded ethanol series and embedded in epoxy resin. Ultrathin (80 nm) sections were prepared, mounted on copper grids, and double-stained with uranyl acetate and lead citrate. Images were captured using a Hitachi HT-7800 transmission electron microscope (Tokyo, Japan) at 120 kV. Multiple random fields were observed for each biological replicate.

2.7 Statistical analysis

All statistical analyses were performed using SPSS 26.0 (IBM, USA) and OriginPro 2024b (OriginLab, USA), with differences considered statistically significant at $P < 0.05$. Data were first checked for normality and homogeneity of variance using the Shapiro–Wilk and Levene's tests, respectively. In the figures, data are presented as mean ± standard error of the mean (SEM), whereas in the supplementary tables, they are presented as mean ± standard deviation (SD).

A three-way analysis of variance (ANOVA) was conducted to evaluate the main and interactive effects of sex, exposure concentration, and exposure duration on each biochemical and molecular index. When significant effects were detected, one-way ANOVA followed by Tukey's honestly significant difference (HSD) post-hoc test was performed to further compare differences among treatment groups at each time point within each sex, as well as differences among time points within each treatment group. Correlation analysis among all biochemical indicators was performed using Pearson's correlation coefficient, and principal component analysis (PCA) was conducted to visualize the overall differences between male and female mussels.

3. Results

3.1 MC-LR accumulation in the gonadal tissues

The temporal accumulation of MC-LR in the gonads of female and male mussels exposed to varying environmental levels is shown in **Fig 1**. Low but detectable MC-LR concentrations were observed in the control group across all sampling points. No significant temporal variation was detected within the control group during the exposure period. A three-way ANOVA revealed significant main effects of exposure concentration ($P < 0.05$) and time ($P < 0.05$) on MC-LR accumulation. Moreover, significant interactions were observed between concentration and time ($P < 0.05$) (**Table S2**). At the same exposure concentration, MC-LR content in gonadal tissues increased progressively from 0 d to 4 d ($P < 0.05$). At the same exposure time, the gonadal MC-LR content increased significantly with increasing exposure concentrations ($P < 0.05$). In the 10 μg/L group, MC-LR levels exhibited a nearly linear increase throughout the exposure period. In the 40 μg/L group, MC-LR concentration increased almost linearly from 0 d to 3 d, followed by a slower rate of increase thereafter. In the 70 μg/L group, the accumulation rose rapidly within the first two days and then gradually leveled off, showing a reduced rate of increase between 2 d and 4 d.

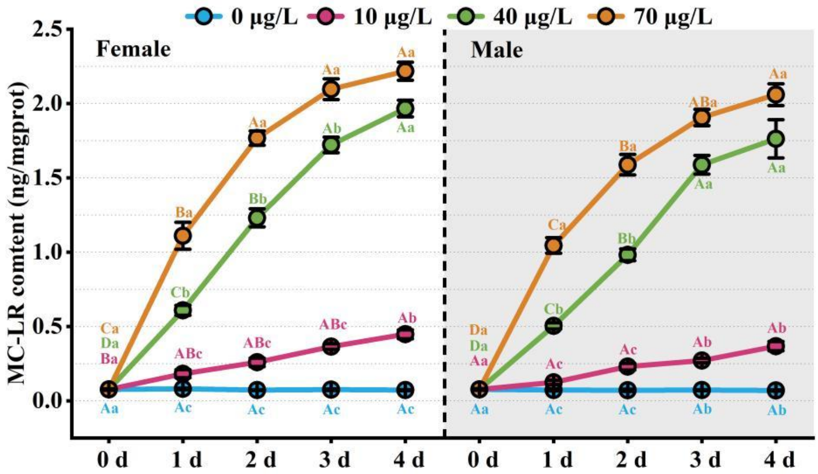

Fig 1. MC-LR accumulation in the gonads of female and male mussels under different exposure concentrations. Tissues were sampled after exposure to 0, 10, 40, and 70 μg/L MC-LR for 0, 1, 2, 3, and 4 d. Data are presented as mean ± SEM (n = 9). Differences among concentrations and exposure times were considered significant at $P < 0.05$. Uppercase letters (A, B, C...) indicate significant differences ($P < 0.05$) among different time points within the same concentration group. Lowercase letters (a, b, c...) indicate significant differences ($P < 0.05$) among different concentration groups within the same time point. Detailed statistical results are provided in **Table S3**.

3.2 Effects of MC-LR exposure on oxidative stress responses in the gonad

The activities of antioxidant enzymes (SOD, CAT), the level of non-enzymatic antioxidant (GSH), and the levels of oxidative stress markers (ROS, MDA) in the gonads of female and male mussels exposed to different MC-LR concentrations are shown in **Fig 2**. A three-way ANOVA revealed significant main effects of exposure concentration ($P < 0.05$), exposure time ($P < 0.05$), and sex ($P < 0.05$) on all measured parameters. Significant interactions were also observed between concentration and time for all indicators ($P < 0.05$), Meanwhile, significant interactions were also found between time and gender, as well as between concentration and gender for ROS and MDA ($P < 0.05$) (**Table S2**).

The activities of both SOD and CAT exhibited a clear biphasic response to MC-LR exposure in both sexes, characterized by an initial induction at lower concentrations and subsequent inhibition at higher concentrations or longer exposure times (**Fig 2A, B**). For both enzymes, exposure to 10 μg/L MC-LR induced a significant increase in activity, peaking at 2 d ($P < 0.01$), before declining to levels comparable with the control group by 4 d. At 40 μg/L, the induction was more rapid, with activity peaking at 1 d ($P < 0.01$), followed by a steep decline to levels significantly below the control group by 3 - 4 d ($P < 0.05$). In contrast, the highest concentration (70 μg/L) caused a direct inhibitory effect, with both SOD and CAT activities becoming significantly lower than in the control group by 2 d ($P < 0.05$). The response patterns were highly comparable between males and females.

Consistent with the impairment of the enzymatic antioxidant system, MC-LR exposure caused a dose- and time-dependent depletion of GSH (**Fig 2C**) and a corresponding accumulation of ROS (**Fig 2D**) and MDA (**Fig 2E**). GSH content showed a sharp decline in the 40 and 70 μg/L exposure groups in both sexes, becoming significantly lower than the control by 1-2 d ($P < 0.05$). In males, the depletion was slightly more severe, with a significant reduction observed even in the 10 μg/L group by 3 d. This depletion of antioxidant capacity was accompanied by a surge in oxidative damage. ROS and MDA levels increased significantly in all MC-LR treatments compared to the control. In the 40 and 70 μg/L groups, this increase was immediate and substantial, evident from 1 d onward ($P < 0.01$). In the 10 μg/L group, the increase was more gradual, with ROS levels peaking at 2 d and MDA levels becoming significant by 3 d. Throughout the experiment, a clear dose-response relationship was observed, with the 70 μg/L group consistently showing the highest levels of ROS and MDA in both sexes.

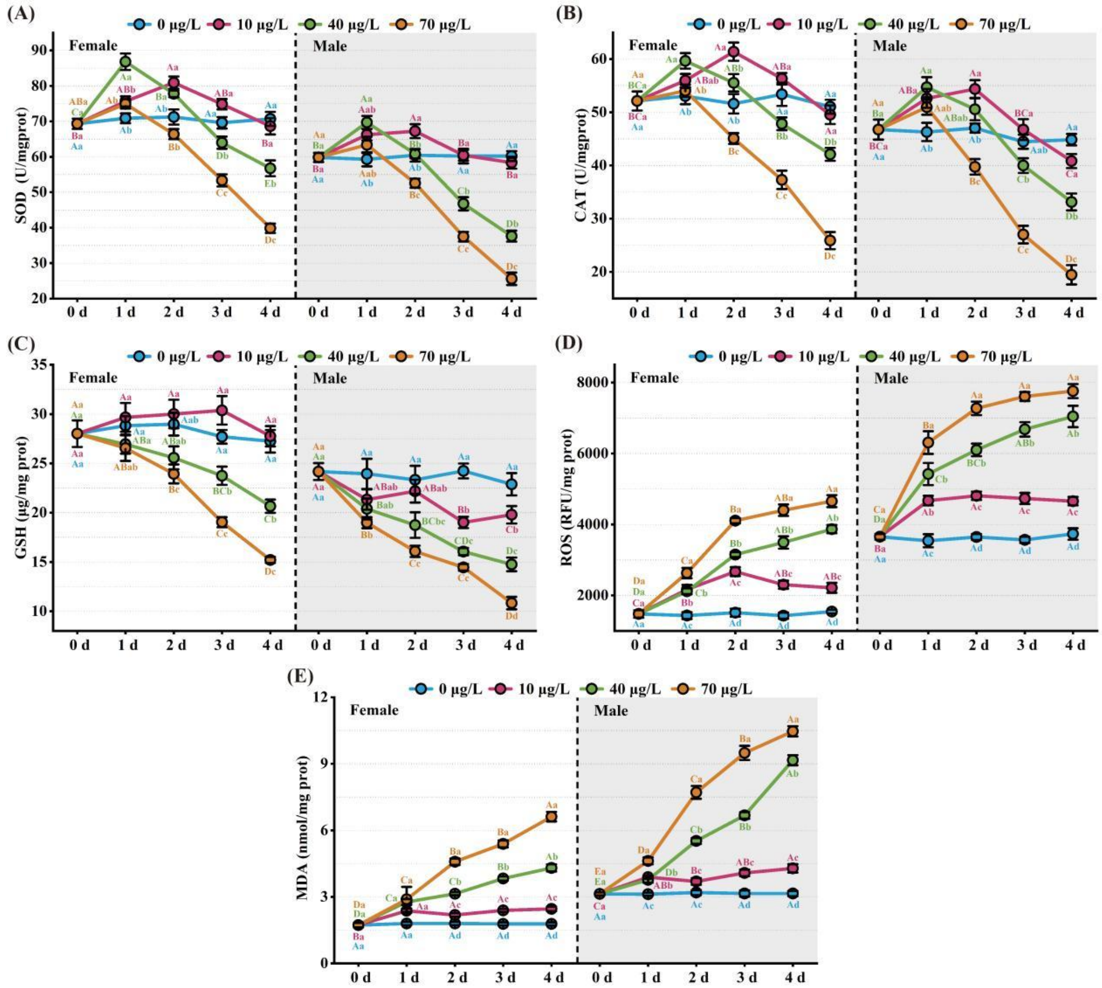

Fig 2. Oxidative stress responses in the gonads of female and male mussels following exposure to different concentrations of MC-LR. The experimental design included four concentrations (0, 10, 40, and 70 μg/L), with tissues sampled after 0, 1, 2, 3, and 4 d of exposure. Panels show: (A) SOD activity, (B) CAT activity, (C) GSH content, (D) ROS levels, and (E) MDA content. Data are presented as mean ± SEM (n = 9). The significance threshold for differences between concentration gradients and exposure times was set at $P < 0.05$. Uppercase letters (A, B, C...) indicate significant differences ($P < 0.05$) among different time points within the same concentration group. Lowercase letters (a, b, c...) indicate significant differences ($P < 0.05$) among different concentration groups within the same time point. Detailed statistical results are provided in **Table S4-8**.

3.3 Mitochondrial dysfunction and ferroptosis indicators in gonads

Exposure to MC-LR induced significant mitochondrial damage and activated key molecular markers of ferroptosis in the gonads of both male and female mussels (**Fig 3**). A three-way ANOVA confirmed that exposure concentration, time, and sex had significant main effects on all measured indicators: $\Delta\Psi_m$, Fe^{2+} levels, and the relative expression of *GPX4*, acyl-CoA synthetase long-chain family member 4 (*ACSL4*), and prostaglandin-endoperoxide synthase 2 (*PTGS2*) ($P < 0.05$). Furthermore, significant interactions were observed between concentration and time for all five parameters ($P < 0.05$). Significant interactions between time and sex were detected for $\Delta\Psi_m$, Fe^{2+}, *GPX4*, and *PTGS2* ($P < 0.05$); significant interactions between concentration and sex were found for $\Delta\Psi_m$, Fe^{2+}, and *PTGS2* ($P < 0.05$), while a three-way interaction (concentration × time × sex) was significant for *PTGS2* (**Table S2**).

MC-LR exposure caused a dose- and time-dependent decrease in $\Delta\Psi_m$, indicating mitochondrial depolarization (**Fig 3A**). In both sexes, the loss of $\Delta\Psi_m$ was most rapid and severe in the highest concentration group (70 μg/L), with a significant decline observed as early as 1 d ($P < 0.01$). The 40 μg/L and 10 μg/L groups also showed significant decreases by 2 d and 3 d, respectively.

Consistent with the induction of ferroptosis, the expression of the key anti-ferroptotic enzyme *GPX4* was significantly downregulated (**Fig 3C**). This suppression was dose- and time-dependent, with the 40 and 70 μg/L groups exhibiting a significant reduction from 1 - 2 d onward in females. In males, the effect was most pronounced at the highest concentration by 4 d.

Concurrently, MC-LR led to a significant, dose-dependent accumulation of intracellular Fe^{2+} (**Fig 3B**) and the upregulation of pro-ferroptotic genes *ACSL4* and *PTGS2* (**Fig 3D, E**). Fe^{2+} levels increased steadily in all treatment groups, with the 40 and 70 μg/L exposures causing a significant rise from 1 d onward in both sexes ($P < 0.01$). Similarly, the expression of both *ACSL4* and *PTGS2* was strongly upregulated in a dose-dependent manner. This induction was most robust in the 70 μg/L group, where expression levels were significantly elevated from 1 - 2 d onward ($P < 0.01$).

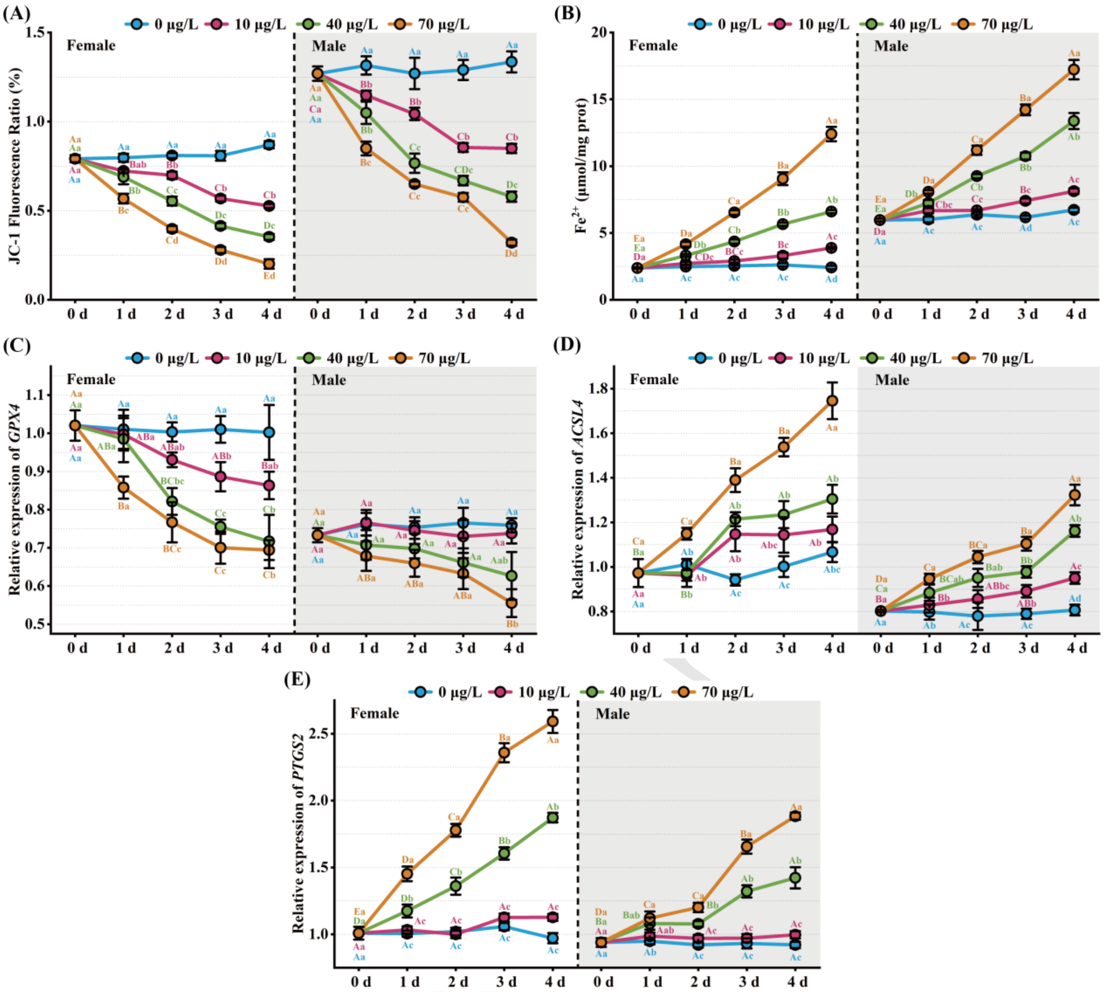

Fig 3. Effects of MC-LR on $\Delta\Psi_m$ and ferroptosis in the gonads of female and male mussels. Mussels were exposed to four concentrations (0, 10, 40, and 70 μg/L) and tissues were sampled after 0, 1, 2, 3, and 4 d. Panels show: (A) $\Delta\Psi_m$; (B) Ferrous iron (Fe^{2+}) content; and relative mRNA expression of (C) *GPX4*, (D) *ACSL4*, and (E) *PTGS2*. Data are presented as mean ± SEM (n = 9). The significance threshold for differences between concentration gradients and exposure times was set at $P < 0.05$. Uppercase letters (A, B, C...) indicate significant differences ($P < 0.05$) among different time points within the same concentration group. Lowercase letters (a, b, c...) indicate significant differences ($P < 0.05$) among different concentration groups within the same time point. Detailed statistical results are provided in **Table S9-13**.

3.4 Ultrastructural changes in gonadal tissues under MC-LR exposure

As shown in **Fig 4**, transmission electron microscopy revealed pronounced ultrastructural alterations in the gonadal mitochondria of mussels after exposure to 70 μg/L MC-LR for 4 d. In the control group, mitochondria in both male and female gonadal cells exhibited an intact double membrane, well-defined cristae, and a dense matrix, suggesting normal mitochondrial function. In contrast, MC-LR exposure led to evident mitochondrial damage. Mitochondria became swollen with disrupted or vesiculated cristae, and some organelles exhibited vacuolar degeneration or loss of membrane integrity. Several mitochondria showed concentric or ring-shaped cristae (red arrows), indicative of severe structural disorganization. In females, although mitochondria generally retained elongated shapes, clear morphological abnormalities were observed, including partial cristae rupture, matrix clearance, and the appearance of electron-lucent areas.

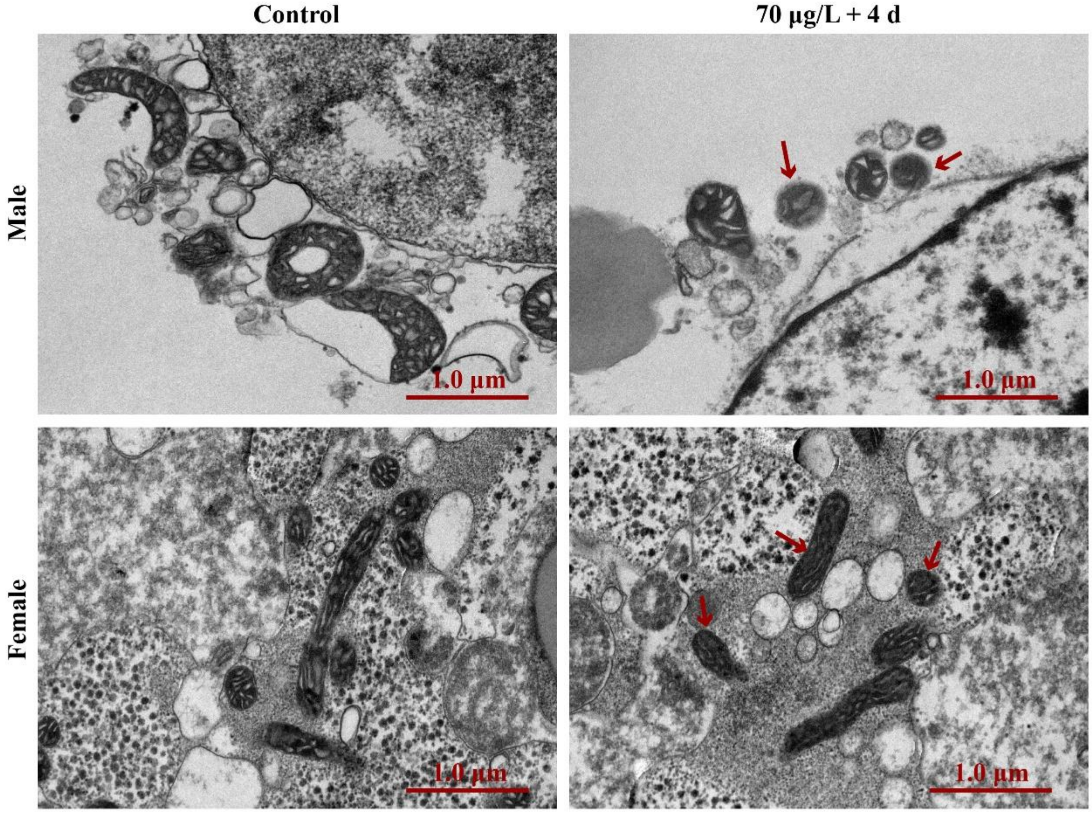

Fig 4. Ultrastructural changes in gonadal mitochondria of mussels after exposure to 70 μg/L MC-LR for 4 d. Representative TEM images show mitochondria in male (top) and female (bottom) gonadal cells. Red arrows indicate severe mitochondrial damage, such as ruptured cristae and condensed matrices, in the MC-LR-exposed groups compared to controls. Scale bars = 1.0 μm.

3.5 Pearson Correlation Analysis and Principal component analysis (PCA)

To further clarify the relationship between oxidative stress, mitochondrial function, and ferroptosis-related indicators in the gonads of mussels under MC-LR exposure, correlation and principal component analyses were performed (**Fig 5**).

Sex-specific correlation matrices are presented in **Fig 5A** (female) and **Fig 5B** (male), and the combined correlation matrix is shown in **Fig 5C**. In both females and males, significant positive correlations were observed among antioxidant enzymes (SOD, CAT, GSH) and *GPX4*, while these parameters were negatively correlated with ROS, MDA, and Fe^{2+} levels. MC-LR, *ACSL4*, and *PTGS2* were positively associated with oxidative damage markers (ROS and MDA), suggesting their involvement in lipid peroxidation and ferroptotic processes. In contrast, JC-1, representing $\Delta\Psi_m$, showed a negative correlation with oxidative and ferroptotic indicators, indicating mitochondrial dysfunction under toxin exposure.

Principal component analysis (PCA) further distinguished clear clustering between male and female gonadal responses (**Fig 5D**). The first two principal components (PC1 = 60.3%, PC2 = 20.0%) explained 80.3% of the total variance. Female samples were primarily associated with elevated antioxidant capacity (SOD, CAT, GSH, *GPX4*), whereas male samples were characterized by higher ROS, MDA, and Fe^{2+} accumulation, as well as increased expression of ferroptosis-related genes (*ACSL4, PTGS2*, MC-LR). Principal component analysis based on exposure duration and MC-LR concentration further illustrated multivariate response patterns among treatment groups (**Fig 5E, F**). Samples from different exposure times and concentration levels showed gradual distribution shifts along the principal component axes; however, considerable overlap among groups was observed in the PCA space.

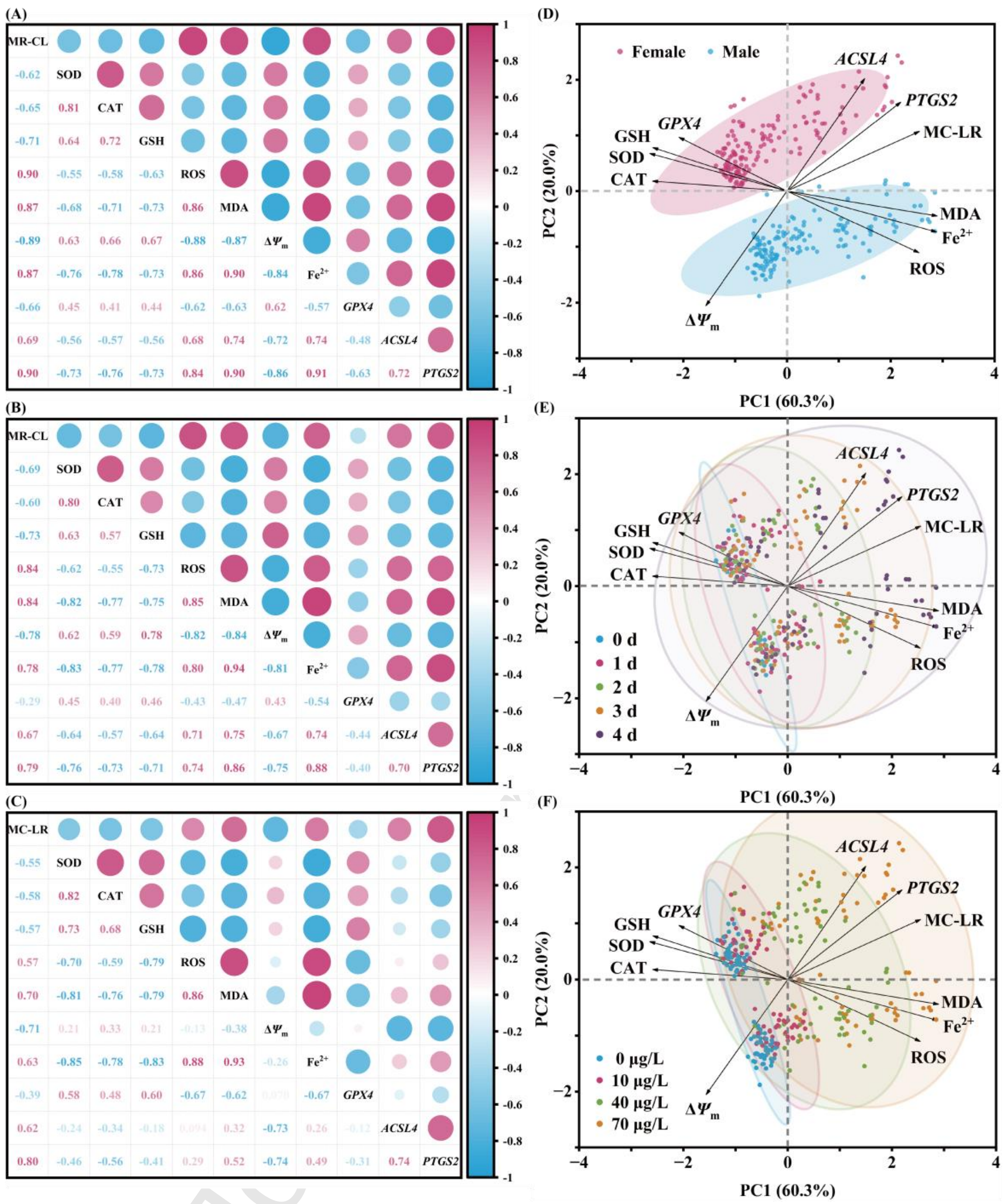

Fig 5. Correlation matrix and principal component analysis (PCA) of oxidative stress and ferroptosis-related parameters in the gonads of mussels exposed to MC-LR. (A) Pearson correlation matrix for females. (B) Pearson correlation matrix for males. (C) Pearson correlation matrix for all samples. The size of the circles is proportional to the correlation strength, while the color indicates the direction of the correlation (pink = positive, blue = negative). (D) PCA biplot of sample scores and variable loadings grouped by sex. (E) PCA biplot of sample scores grouped by exposure duration (0, 1, 2, 3 and 4 d). (F) PCA biplot of sample scores grouped by MC-LR concentration (0, 10, 40 and 70 μg/L). Samples are colored as female (pink) and male (blue), and

ellipses indicate the 95% confidence intervals for each group.

4. Discussion

Against the background of a global decline in freshwater mussel populations [25], elucidating the toxicological mechanisms by which pollutants impair their reproductive systems is of critical importance. Although the reproductive toxicity of MC-LR has been well-documented [26, 27], the specific mode of cell death and the underlying mechanisms of its sex-specific effects remain unclear. This study provides the first systematic evidence that environmentally relevant concentrations of MC-LR induce reproductive toxicity in the gonads of the mussels by triggering ferroptosis. Furthermore, we reveal that males exhibit a higher susceptibility (**Fig 6A**), offering a new perspective on the toxicology of MC-LR.

MC-LR accumulated in the gonadal tissue of mussels in a time- and concentration-dependent manner. The accumulation rate was rapid during the initial phase of exposure and subsequently plateaued, a trend likely attributable to the progressive saturation of cell surface binding sites and the activation of metabolic and excretory pathways [28]. Unlike primary detoxification organs such as the hepatopancreas, gonadal tissue has a limited capacity to metabolize and clear MC-LR, owing to lower levels of detoxification enzymes like glutathione S-transferase (GST) [29, 30]. This imbalance between high exposure and low clearance led to the continuous bioaccumulation of MC-LR in the gonads, which directly triggered the subsequent oxidative stress and cellular damage observed in our study.

Oxidative stress is considered a primary mechanism of MC-LR toxicity [31, 32]. MC-LR exposure triggers the excessive generation of reactive oxygen species (ROS) in vivo, which are typically neutralized by major antioxidant enzymes such as SOD and CAT [33-35]. In this study, the antioxidant system in the gonads of the mussels exhibited a dynamic process, transitioning from compensatory activation to eventual collapse. Initially, the activities of SOD and CAT were transiently elevated, representing a typical stress defense response aimed at scavenging excess ROS. However, with prolonged toxin exposure, the levels of reactive oxygen species (ROS) and malondialdehyde (MDA) increased significantly, accompanied by a sharp decline in enzymatic activities. The mitochondria and endoplasmic reticulum are considered the primary sources of MC-LR-induced ROS, whose products can further oxidize cellular membrane lipids, proteins, and DNA [36]. As the oxidative defense system gradually loses balance [37], enzyme inactivation under high oxidative stress or direct inhibition by MC-LR may occur [38], a phenomenon also observed in *Diplodon chilensis patagonicus* [*39*] and *Corbicula fluminea* [40]. Concomitantly, the persistent depletion of the nonenzymatic antioxidant GSH further aggravates redox imbalance. GSH is not only extensively consumed for direct ROS neutralization and lipid peroxidation repair [41], but its synthesis is also inhibited by MC-LR. Previous studies have demonstrated that MC-LR markedly suppresses the activity of γ-glutamylcysteine ligase (GCL), the rate-limiting enzyme in the GSH biosynthetic pathway [42, 43]. The depletion of GSH represents a turning point, as it not only causes a dramatic accumulation of oxidative damage products such as ROS and MDA, but more importantly, directly weakens the function of the key anti-ferroptotic enzyme GPX4, rendering cells highly susceptible to ferroptosis [44].

As the antioxidant system becomes progressively imbalanced, mitochondria, as the central hub for ROS production and energy metabolism, emerge as a primary target of MC-LR toxicity [45]. The depletion of the GSH–GPX4 defense axis compromises the cell's buffering capacity against oxidative damage, leading to mitochondrial dysfunction and further ROS leakage [46]. In this study, our JC-1 staining results revealed a significant decrease in the $\Delta\Psi_m$ in the gonads of both sexes after MC-LR exposure, indicating a compromised proton gradient and increased membrane permeability [47]. This depolarization process typically signifies an impaired electron transport chain (ETC) and reduced efficiency of ATP synthesis, which in turn amplifies oxidative stress [48]. This mitochondrial depolarization–associated oxidative stress is consistent with previous reports showing that prolonged MC-LR exposure induces mitochondrial swelling, enhanced oxidative phosphorylation activity, and excessive ROS generation in testicular tissue, ultimately leading to reproductive impairment [49]. Similar findings, including decreased $\Delta\Psi_m$

and reduced ATP production following cyanotoxin exposure, have been reported in the central nervous system of *Nile Tilapia* [50] and the lymphocytes of *Carassius auratus* [51]. Transmission electron microscopy (TEM) observations provided direct ultrastructural evidence for this functional impairment. Mitochondria in the exposed groups exhibited typical morphological changes, including swelling, rupture of cristae, and increased matrix density. These structural damages are considered characteristic hallmarks of both oxidative stress and iron-dependent cell death [24].

Ferroptosis is a regulated form of cell death characterized by excessive iron accumulation and lipid peroxidation [52]. The sustained oxidative stress and mitochondrial dysfunction induced by MC-LR create favorable conditions for its initiation [53]. In this study, increasing MC-LR concentrations resulted in a significant elevation of Fe^{2+} levels in gonadal tissues, accompanied by a concomitant decline in GSH content and *GPX4* transcriptional levels. These findings suggest that MC-LR may trigger ferroptosis by disrupting the cellular antioxidant defense system and promoting intracellular iron enrichment. Recent studies have further implicated ferroptosis as an important mechanism underlying microcystin-induced toxicity. For instance, transcriptomic analysis in a rat model demonstrated that microcystin exposure activated ferroptosis-related pathways associated with oxidative stress imbalance and metal ion dysregulation during the development of hepatic encephalopathy [54]. Excess Fe^{2+} can catalyze hydroxyl radical formation through the Fenton reaction, which subsequently oxidizes polyunsaturated fatty acids (PUFAs) within membrane lipids, leading to membrane rupture and cell death [55]. GPX4 serves as the central defense enzyme against ferroptosis by utilizing GSH as a reducing substrate to convert toxic lipid hydroperoxides (L-OOH) into harmless lipid alcohols (L-OH), thereby blocking iron-dependent oxidative chain reactions [56, 57]. Thus, MC-LR–induced GSH depletion and *GPX4* inactivation jointly compromise the cellular lipid peroxidation barrier and accelerate ferroptotic progression. Furthermore, ACSL4 and PTGS2, two canonical molecular markers of ferroptosis [58, 59], were markedly upregulated following exposure. ACSL4 facilitates PUFA acylation, providing essential substrates for lipid peroxidation [60], whereas elevated PTGS2 expression reflects enhanced lipid oxidative damage and inflammatory responses [61]. The concurrent upregulation of ACSL4 and PTGS2 further confirms that MC-LR exposure induces a typical ferroptotic response, highlighting a continuous toxicological cascade from oxidative stress and mitochondrial dysfunction to ferroptotic cell death.

One of the most noteworthy findings of this study is the marked sex-dependent difference in MC-LR–induced ferroptosis, with males exhibiting higher susceptibility. Sex-dependent effects of MC-LR have also been reported in other animal models. In fish, several studies have demonstrated that MC-LR exposure induces differential reproductive and cellular responses between males and females. For example, in *Danio rerio* [62] and *Oryzias latipes* [63], chronic or subchronic MC-LR exposure resulted in more pronounced alterations in reproductive parameters, metabolic processes, and tissue damage in females compared to males. These findings indicate that sex-dependent susceptibility to MC-LR is a common toxicological feature across aquatic vertebrates.

The mussel, as a bivalve species possessing a DUI system, maintains two distinct mitochondrial lineages (F-type in females and M-type in males) (**Fig 6B**). This unique genetic structure provides a molecular foundation for sex-specific responses to environmental toxic stress. The PCA results revealed clear sexual dimorphism in the gonadal responses of mussels following MC-LR exposure, indicating that males and females adopt distinct strategies in toxin uptake, antioxidant defense, and metabolic regulation. The accumulation of MC-LR in female gonads was notably higher than in males, which may be attributed to their higher lipid content and active reproductive metabolism. During oogenesis, females synthesize large amounts of lipids and steroid hormones, requiring substantial energy input, while MC-LR, with its lipophilic nature, can easily bind to membrane lipids or intracellular proteins to form stable complexes [64-66]. Meanwhile, correlation analysis showed that in females, antioxidant factors (SOD, CAT, GSH, *GPX4*) were positively correlated with lipid metabolism–related genes (*ACSL4*, *PTGS2*), suggesting that females possess stronger redox regulatory capacity. Such a more robust antioxidant defense system may enable females to tolerate prolonged toxic stress more effectively. In contrast, male samples showed strong positive correlations with oxidative damage indicators such as ROS, Fe^{2+}, and MDA, and maintained a relatively higher $\Delta\Psi_m$ compared to females, despite an overall decline following MC-LR exposure. This suggests a sex-dependent bioenergetic background rather than an exposure-induced enhancement of ETC activity. Although a higher $\Delta\Psi_m$ favors ATP synthesis, it also increases the risk of electron leakage and ROS generation, rendering males more vulnerable to lipid peroxidation and mitochondrial injury [67]. This pattern is consistent with the physiological characteristics associated with male gonadal mitochondria under the doubly uniparental inheritance (DUI) system. Previous studies on DUI bivalves have demonstrated intrinsic bioenergetic differences between sperm-derived and oocyte-derived mitochondria. For example, in species such as *Mytilus edulis*, *Ruditapes philippinarum*, and *Arctica islandica* [8,9], mitochondrial complex IV activity in sperm has been reported to be significantly lower than that in oocytes, indicating reduced oxidative phosphorylation coupling efficiency. Such reduced coupling does not enhance ATP production; instead, it is often associated with alterations in $\Delta\Psi_m$ and an increased probability of electron leakage from the electron transport chain. This bioenergetic state favors elevated ROS generation and places male gonadal mitochondria in a more oxidatively stressed condition [5,68]. Under toxic challenges such as MC-LR exposure, this inherent redox vulnerability may predispose male gonadal tissues to enhanced lipid peroxidation, mitochondrial dysfunction, and ferroptosis-related damage.

Against the backdrop of a global decline in freshwater bivalve populations, the sex-specific toxicological mechanisms revealed in this study carry important ecological implications. MC-LR–induced ferroptosis poses a direct threat to gonadal function, particularly in males, and may therefore reduce reproductive capacity and alter population structure. In species exhibiting DUI, such male-biased vulnerability could further disrupt the balance of mitochondrial transmission and compromise long-term population stability. It should be noted that the present study focuses on acute to sub-acute exposure, which is appropriate for identifying early mechanistic events but does not fully represent chronic, low-level environmental exposure scenarios. Under long-term conditions, compensatory responses and cumulative reproductive effects may occur. Future studies incorporating prolonged, environmentally relevant exposures across reproductive cycles will be necessary to improve ecological risk assessment. In the context of increasingly frequent cyanobacterial blooms driven by global climate change, a deeper understanding of sex-specific responses to pollutants is crucial for accurately assessing ecological risks and developing effective conservation strategies for freshwater mollusks and other vulnerable aquatic species.

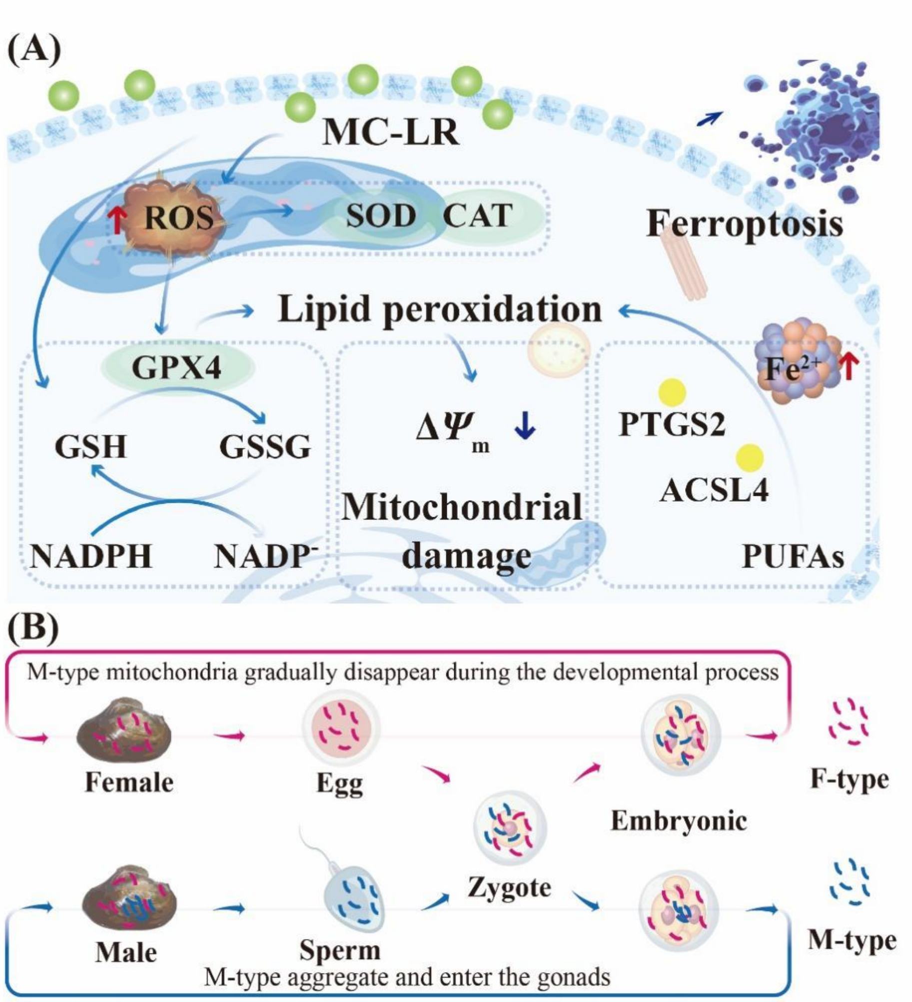

Fig 6. Schematic diagram illustrating the proposed mechanism of MC-LR-induced gonadal ferroptosis (A) and the DUI system in the mussel (B). (A) MC-LR exposure induces excessive ROS production and mitochondrial dysfunction in gonadal tissues, leading to antioxidant system disruption characterized by GSH depletion and GPX4 inhibition. Concurrently, intracellular Fe^{2+} accumulation promotes lipid peroxidation

through iron-dependent oxidative reactions, resulting in membrane damage and activation of ferroptosis-related pathways, including upregulation of ACSL4 and PTGS2. These processes collectively contribute to ferroptotic cell injury in mussel gonads. (B) The DUI system in bivalves is characterized by the coexistence of maternally inherited mitochondrial DNA (F-type) and paternally transmitted mitochondrial DNA (M-type), which are differentially distributed between female and male gonadal tissues. This unique mitochondrial inheritance pattern may underlie sex-specific differences in mitochondrial function and susceptibility to MC-LR-induced oxidative stress and ferroptosis.

CRediT authorship contribution statement

Liusiqiao Tang: Formal analysis, Methodology, Software, Visualization, Writing – review & editing. Xuechun Wang: Writing - Original draft preparation, Formal analysis, Investigation, Software. Yang Gu: Writing - Original draft preparation, Formal analysis, Data curation. Jianwei Wang: Investigation, Visualization. Yingrui Mao: Investigation, Visualization. Qinyun Hu: Visualization. Guiling Wang: Conceptualization, Methodology, Funding acquisition, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by National Natural Science Foundation of China (grant no. 32573497), the National Key R&D Program of China (2022YFD2400700), the earmarked fund for CARS (CARS-49).

Reference

[1] K. C. Cushway, J. Geist, A. N. Schwalb, Surviving global change: a review of the impacts of drought and dewatering on freshwater mussels, Biol. Rev. 100 (1) (2025) 275-307, https://doi.org/10.1111/brv.13142

[2] W. R. Haag, J. D. Williams, Biodiversity on the brink: an assessment of conservation strategies for North American freshwater mussels, Hydrobiologia. 735 (1) (2014) 45-60, https://link.springer.com/article/10.1007/s10750-013-1524-7.

[3] M. Böhm, N. I. Dewhurst-Richman, M. Seddon, S. E. Ledger, C. Albrecht, D. Allen, A. E. Bogan, J. Cordeiro, K. S. Cummings, A. J. H. Cuttelod, The conservation status of the world’s freshwater molluscs, Hydrobiologia. 848 (12) (2021) 3231-3254, https://www.researchgate.net/publication/344236160.

[4] D. C. Aldridge, I. S. Ollard, Y. V. Bespalaya, I. N. Bolotov, K. Douda, J. Geist, W. R. Haag, M. W. Klunzinger, M. Lopes-Lima, M. C. Mlambo, Freshwater mussel conservation: A global horizon scan of emerging threats and opportunities, Glob. Chang. Biol. 29 (3) (2023) 575-589, https://doi.org/10.1111/gcb.16510.

[5] A. Pouliot-Drouin, T. Niaison, S. Breton, S. Bettinazzi, Investigating the role of mitochondrial membrane potential in paternal inheritance of mitochondria, Biol. J. Linn. Soc. 144(3) (2025) blae050, https://doi.org/10.1093/biolinnean/blae050.

[6] C. H. Smith, Evolution. A high-quality reference genome for a parasitic bivalve with doubly uniparental inheritance (Bivalvia: Unionida), Genome Biol. Evol. 13 (3) (2021) evab029, https://doi.org/10.1093/gbe/evab029.

[7] C. H. Smith, R. Mejia-Trujillo, J. C. Havird, Mitonuclear compatibility is maintained despite relaxed selection on male mitochondrial DNA in bivalves with doubly uniparental inheritance, Evol. 78 (11) (2024) 1790-1803, https://doi.org/10.1093/evolut/qpae108.

[8] A. Gusman, S. Lecomte, D. T. Stewart, M. Passamonti, S. Breton, Pursuing the quest for better understanding the taxonomic distribution of the system of doubly uniparental inheritance of mtDNA, PeerJ. 4 (2016) e2760, https://peerj.com/articles/2760/.

[9] S. Bettinazzi, L. Milani, P. U. Blier, S. Breton, Bioenergetic consequences of sex-specific mitochondrial DNA evolution, Proc. R. Soc. B. 288 (1957) (2021) 20211585, https://doi.org/10.1098/rspb.2021.1585.

[10] S. Bettinazzi, E. Rodríguez, L. Milani, P. U. Blier, S. Breton, Metabolic remodelling associated with mtDNA: insights into the adaptive value of doubly uniparental inheritance of mitochondria, Proc. R. Soc. B. 286 (1896) (2019) 20182708, https://doi.org/10.1098/rspb.2018.2708.

[11] M. Yan, C. Wang, H. Wu, T. Wu, L. Fang, X. Han, Screening, identification and functional validation of Microcystin-LR direct binding target proteins based on thermal proteomics profiling, *Sci. Total Environ. 958* (2025) 178047.

[12] B. Li, Y. Liu, Y. Liu, P. Xie, Excluding interference and detecting Microcystin-LR in the natural lakes and cells based a unique fluorescence method, Water Res. 221 (2022) 118811. https://doi.org/10.1016/j.watres.2022.118811.

[13] A. Hao, D. Xia, X. Mou, S. Kobayashi, T. Haraguchi, Y. Iseri, M. Zhao, Bacterial Communities Respond to Spatiotemporal Fluctuation in Water Quality and Microcystins at Lake Taihu, Water. 17 (15) (2025) 2222, https://doi.org/10.3390/w17152222.

[14] H. Chuan, B. Li, Z. Wang, Y. Zhang, P. Xie, Y. Liu, Visualization of MC-LR in lakes using near-infrared technology, Water Res. 281 (2025) 123558, https://doi.org/10.1016/j.watres.2025.123558

[15] H. Zhang, X. Zhao, Y. Li, P. Xie, A meta-analysis on the toxicity of microcystin-LR to fish and mammals, Environ. Pollut. 330 (2023) 121780, https://doi.org/10.1016/j.envpol.2023.121780.

[16] Y. Liu, M. Yang, L. Zheng, H. Nguyen, L. Ni, S. Song, Y. Sui, Antioxidant responses of triangle sail mussel *Hyriopsis cumingii* exposed to toxic *Microcystis aeruginosa* and thermal stress, Sci. Total Environ. 2020, 743, 140754, https://doi.org/10.1016/j.scitotenv.2020.140754.

[17] J. Wu, S. Shao, F. Zhou, S. Wen, F. Chen, X. Han, Reproductive toxicity on female mice induced by microcystin-LR, Environ. Toxicol. Pharmacol. *37*(1) (2014) 1-6. https://doi.org/10.1016/j.etap.2013.10.012.

[18] J. Hou, Y. Su, W. Lin, H. Guo, P. Xie, J. Chen, Z. Gu, L. Li, (2017). Microcystin-LR retards gonadal maturation through disrupting the growth hormone/insulin-like growth factors system in zebrafish, Ecotoxicol. Environ. Saf. 139 (2017) 27-35. https://doi.org/10.1016/j.ecoenv.2017.01.025.

[19] Z. Liu, Y. Zhang, X. Jia, T. D. Hoskins, L. Lu, Y. Han, X. Zhang, H. Lin, L. Shen, Y. Feng, Y. Zheng, C. Hu, H. Zhang, Microcystin-LR induces estrogenic effects at environmentally relevant concentration in black-spotted pond frogs (*Pelophylax nigromaculatus*): *in situ*, *in vivo*, *in vitro*, and *in silico* investigations, Environ. Sci. Technol. 2024 58 (22) 9559-9569. https://doi.org/10.1021/acs.est.4c01322

[20] Y. Zhang, H. Zhuang, H. Yang, W. Xue, L. Wang, W. Wei, Microcystin-LR disturbs testicular development of giant freshwater prawn *Macrobrachium rosenbergii, Chemosphere 222* (2019) 584-592. https://doi.org/10.1016/j.chemosphere.2019.01.146.

[21] P. Yu, L. Su, B. Li, J. Su, G. Yuan, Selenomethionine alleviates Aeromonas hydrophila-induced oxidative stress and ferroptosis via the Nrf2/HO1/GPX4 pathway in grass carp, Fish Shellfish Immunol. 154 (2024) 109927, https://doi.org/10.1016/j.fsi.2024.109927.

[22] K. Cheng, G. Yang, M. Huang, Y. Huang, C. Wang, Exogenous 1, 25 (OH) $_2D_3$/VD_3 counteracts RSL3-Induced ferroptosis by enhancing antioxidant capacity and regulating iron ion transport: Using zebrafish as a model, Chem. Biol. Interact. 387 (2024) 110828, https://doi.org/10.1016/j.cbi.2023.110828.

[23] S. C. Li, L. H. Gu , Y. F. Wang, L.M. Wang, L. Chen, J. Giesy, X. Tuo, W. L. Xu, Q. H. Wu, Y. Q. Liu, M. H. Wu, Y. Y. Diao, H. H. Zeng, Q. B. Zhang, A proteomic study on gastric impairment in rats caused by microcystin-LR, Sci. Total Environ. 917 (2024) 169306, https://doi.org/10.1016/j.scitotenv.2023.169306

[24] Y. Zhou, W. Liu, H. Jiang, F. Chen, Y. Li, J. Gardea-Torresdey, X. Zhou, Bi. Yan, Surface-charge-driven ferroptosis and mitochondrial dysfunction is involved in toxicity diversity in the marine bivalve exposed to nanoplastics, ACS Nano. 18 (3) (2024) 2370-2383, https://doi.org/10.1021/acsnano.3c10536.

[25] M. Lopes-Lima, R. Sousa, J. Geist, D. C. Aldridge, R. Araujo, J. Bergengren, Y.

Bespalaya, E. Bódis, L. Burlakova, D. Van Damme, K. Douda, E. Froufe, D. Georgiev, C. Gumpinger, A. Karatayev, Ü. Kebapçi, I. Killeen, J. Lajtner, B.M. Larsen, R. Lauceri, A. Legakis, S. Lois, S. Lundberg, E. Moorkens, G. Motte, K. Nagel, P. Ondina, A. Outeiro, M. Paunovic, V. Prié, T. Proschwitz, N. Riccardi, M. Rudzīte, M. Rudzītis, C. Scheder, M. Seddon, H. Şereflişan, V. Simić, S. Sokolova, K. Stoeckl, J. Taskinen, A. Teixeira, F. Thielen, T. Trichkova, S. Varandas, H. Vicentini, K. Zajac, T. Zajac, S. Zogaris, Conservation status of freshwater mussels in Europe: state of the art and future challenges, Biol. Rev. 92 (1) (2017) 572-607, https://doi.org/10.1111/brv.12244.

[26] A. G. Boegehold, N. S. Johnson, J. L. Ram, D. R. Kashian, Cyanobacteria reduce quagga mussel (*Dreissena rostriformis bugensis*) spawning and fertilization success, Freshw. Sci. 37 (3) (2018) 510-518, https://www.journals.uchicago.edu/doi/abs/10.1086/698353.

[27] D. Boltovskoy, N. Correa, F. Bordet, V. Leites, D. Cataldo, Toxic Microcystis (cyanobacteria) inhibit recruitment of the bloom-enhancing invasive bivalve *Limnoperna fortunei*, Freshw. Biol. 58 (9) (2013) 1968-1981, https://doi.org/10.1111/fwb.12184.

[28] M. S. Kim, Y. Lee, S. Y. Ha, B. H. Kim, S. J. Hwang, J. T. Kwon, J. W. Choi, K. H. Shin, Accumulation of microcystin (LR, RR and YR) in three freshwater bivalves in *Microcystis aeruginosa* bloom using dual isotope tracer, Mar. Drugs. 15 (7) (2017) 226, https://doi.org/10.3390/md15070226.

[29] Z. Hong, X. Chen, J. Hu, X. Chang, Y. Qian, Adverse effects of *Microcystis aeruginosa* exudates on the filtration, digestion, and reproduction organs of benthic bivalve *Corbicula fluminea,* Sci. Rep. 14 (1) (2024) 10934. https://www.nature.com/articles/s41598-024-61333-7

[30] G. Le Pennec, M. Le Pennec, Evaluation of the toxicity of chemical compounds using digestive acini of the bivalve mollusc *Pecten maximus L.* maintained alive in vitro, Aquat. Toxicol. 53 (1) (2001) 1-7, https://doi.org/10.1016/S0166-445X(00)00163-6.

[31] Y. He, K. Ouyang, H. Yang, L. Wang, Q. Zhang, D. Li, L. Li, Immunology, S.

The MC-LR induced neuroinflammation and the disorders of neurotransmitter system in zebrafish (*Danio rerio*): Oxidative stress as a key, *Fish Shellfish Immunol.* 2025, *158*, 110126.

[32] L. Cao, N. Shao, J. Du, H. Zhu, J. Gao, Q. Li, Y. Sun, J. Hu, G. Yin, G. Xu, Involvement of reactive oxygen species (ROS) in the hepatopancreatic cytotoxicity, oxidative stress, and apoptosis induced by microcystin-LR in *Eriocheir sinensis*, Comp. Biochem. Physiol. C. 2024, 276, 109801.

[33] W. Lin, H. Guo, L. Wang, D. Zhang, X. Wu, L. Li, D. Li, R. Tang, Nitrite enhances MC-LR-induced changes on splenic oxidation resistance and innate immunity in male zebrafish, Toxins. 10 (12) (2018) 512, https://doi.org/10.3390/toxins10120512.

[34] C. Chi, X. W. Yu, C. Y. Zhang, J. D. Liu, M. W. Ye, D. D. Zhang, W. B. Liu, Acute exposure to microcystin-LR induces hepatopancreas toxicity in the Chinese mitten crab (*Eriocheir sinensis*), Arch. Toxicol. *95* (7) (2021) 2551-2570, https://link.springer.com/article/10.1007/s00204-021-03061-9.

[35] J. Yang, Z. Zhang, X. Du, Y. Wang, R. Meng, K. Ge, C. Wu, X. Liang, H. Zhang, H. Guo, The effect and mechanism of combined exposure of MC-LR and $NaNO_2$ on liver lipid metabolism, Environ. Res. 252 (2024) 119113, https://doi.org/10.1016/j.envres.2024.119113.

[36] D. Zhang, W. Lin, Y. Liu, H. Guo, L. Wang, L. Yang, L. Li, D. Li, R. Tang, Chronic microcystin-LR exposure induces abnormal lipid metabolism via endoplasmic reticulum stress in male zebrafish, Toxins. 12 (2) (2020) 107, https://doi.org/10.3390/toxins12020107.

[37] Y. D. Kim, W. J. Kim, Y. K. Shin, D. H. Lee, Y. J. Kim, J. K. Kim, J. S. Rhee, Microcystin-LR bioconcentration induces antioxidant responses in the digestive gland of two marine bivalves *Crassostrea gigas* and *Mytilus edulis*, Aquat. Toxicol. 188 (2017) 119-129, https://doi.org/10.1016/j.aquatox.2017.05.003.

[38] J. Jiang, Y. Shi, Z. Shan, L. Yang, X. Wang, L. Shi, Bioaccumulation, oxidative stress and HSP70 expression in *Cyprinus carpio L.* exposed to microcystin-LR under laboratory conditions, Comp. Biochem. Physiol. C. 155 (3) (2012) 483-490,

https://doi.org/10.1016/j.cbpc.2011.12.008.

[39] L. Sabatini, B. M. Brena, C. M. Luquet, M. San Julian, M. Pirez, M. d. C. R. de Molina, Microcystin accumulation and antioxidant responses in the freshwater clam *Diplodon chilensis* patagonicus upon subchronic exposure to toxic *Microcystis aeruginosa*, Ecotoxicol. Environ. Saf. 74 (5) (2011) 1188-1194, https://doi.org/10.1016/j.ecoenv.2011.03.012.

[40] J. Zhang, M. Yu, Y. Gao, M. Zhang, J. Dong, M. Li, X. Li, Feeding behavior, microcystin accumulation, biochemical response, and ultramicrostructure changes in edible freshwater bivalve *Corbicula fluminea* exposed to *Microcystis aeruginosa*, Environ. Sci. Pollut. Res. 30 (5) (2023) 13560-13570, https://link.springer.com/article/10.1007/s11356-022-22833-z.

[41] M. Hu, F. Wu, M. Yuan, Q. Li, Y. Gu, Y. Wang, Q. Liu, Antioxidant responses of triangle sail mussel *Hyriopsis cumingii* exposed to harmful algae *Microcystis aeruginosa* and hypoxia, Chemosphere. 139 (2015) 541-549, https://doi.org/10.1016/j.chemosphere.2015.07.074.

[42] L. Chen, P. Xie, Mechanisms of microcystin-induced cytotoxicity and apoptosis, Mini-Rev. Med. Chem. 16 (13) (2016) 1018-1031, https://www.ingentaconnect.com/content/ben/mrmc/2016/00000016/00000013/art00002.

[43] B. Reis, M. Carneiro, J. Machado, J. Azevedo, V. Vasconcelos, J. C. Martins, Transcriptional responses of glutathione transferase genes in *Ruditapes philippinarum* exposed to microcystin-LR, Int. J. Mol. Sci. 16 (4) (2015) 8397-8414, https://doi.org/10.3390/ijms16048397.

[44] H. Ying, H. Zhao, X. Zhang, H. Feng, M. Zhou, Z. Wu, N. Wu, Unraveling the Role of Microcystin-LR in Ferroptosis and Sepsis Pathogenesis: A Comprehensive Review, Biomolecules. 15 (7) (2025) 947, https://doi.org/10.3390/biom15070947.

[45] X. Li, Q. Zhao, W. Zhou, L. Xu, Y. Wang, Effects of chronic exposure to microcystin-LR on hepatocyte mitochondrial DNA replication in mice, Environ. Sci. Technol. 49 (7) (2015) 4665-4672, https://doi.org/10.1021/es5059132.

[46] T. M. Seibt, B. Proneth, Conrad, M. Role of GPX4 in ferroptosis and its

pharmacological implication, Free Radic Biol Med. 133 (2019) 144-152, https://doi.org/10.1016/j.freeradbiomed.2018.09.014

[47] I. Sokolova, S. Evans, F. M. Hughes, Cadmium-induced apoptosis in oyster hemocytes involves disturbance of cellular energy balance but no mitochondrial permeability transition, J. Exp. Biol. 207 (19) (2004) 3369-3380, https://doi.org/10.1242/jeb.01152.

[48] F. Wu, E. P. Sokolov, S. Timm, I. M. Sokolova, Synergistic impacts of nanopollutants (nZnO) and hypoxia on bioenergetics and metabolic homeostasis in a marine bivalve *Mytilus edulis*, Environ. Sci.: Nano. 12 (1) (2025) 576-596, https://doi.org/10.1039/D4EN00479E.

[49] L. Chen, X. Zhang, W. Zhou, Q. Qiao, H. Liang, G. Li, J. Wang, F. Cai, The interactive effects of cytoskeleton disruption and mitochondria dysfunction lead to reproductive toxicity induced by microcystin-LR, PLoS One, 8(1) (2013) e53949. https://doi.org/10.1371/journal.pone.0053949

[50] M. Nájera-Martínez, G. G. Landon-Hernández, J. P. Romero-López, M. L. Domínguez-López, A. Vega-López, Disruption of neurotransmission, membrane potential, and mitochondrial calcium in the brain and spinal cord of nile tilapia elicited by microcystis aeruginosa extract: An uncommon consequence of the eutrophication process, Water, Air, & Soil Pollution. 233 (1) (2022) 6, https://link.springer.com/article/10.1007/s11270-021-05480-x.

[51] K. Zhang, Q. Xiao, J. Jia, Y. Gao, Y. Zhou, X. X. Zhou, B.Yan, Mechanistic Insights into Cadmium Sulfide Nanoparticles-Induced Digestive Gland Damage in *Corbicula fluminea*: Comparison with Cadmium Ions, Environ. Sci. Technol. 59 (17) (2025) 8451-8460. https://doi.org/10.1021/acs.est.5c01150.

[52] H. Zhang, J. Zhang, Y. Chen, Y. Zhu, Microcystin-RR induces apoptosis in fish lymphocytes by generating reactive oxygen species and causing mitochondrial damage, Fish Physiol. Biochem. 34 (4) (2008) 307-312, https://link.springer.com/article/10.1007/s10695-007-9189-7.

[53] G. H. Chen, C. C. Song, K. Pantopoulos, X. L. Wei, H. Zheng, Z. Luo, Mitochondrial oxidative stress mediated Fe-induced ferroptosis via the NRF2-ARE

pathway, Free Radic. Biol. Med. 180 (2022) 95-107, https://doi.org/10.1016/j.freeradbiomed.2022.01.012.

[54] W. L. Xu, L. Chen, H. H. Fan, G. A. Codd, J. P. Giesy, Y. M. Guo, E. D. Hilborn, Daniela Sedan, D. Andrinolo, S. Chatterjee, H. Wang, Y. Liu, Q. Wu, J. He, S. Dai, L. Xu, S. Xiao, Y. Liu, R. Yang, J. Chen, P. Xie, Novel insights into cyanobacterial (microcystins) neurotoxicity in rats: hepatic encephalopathy, Arch. Toxicol. (2025) 1-18, https://doi.org/10.1007/s00204-025-04219-5.

[55] L. Yang, X. Cai, R. Li, Ferroptosis induced by pollutants: an emerging mechanism in environmental toxicology, Environ. Sci. Technol. 58 (5) (2024) 2166-2184, https://doi.org/10.1021/acs.est.3c06127.

[56] S. M. M. Hlaing, J. Lou, J. Cheng, X. Xun, M. Li, W. Lu, X. Hu, Z. Bao, Tissue-biased and species-specific regulation of glutathione peroxidase (*GPx*) genes in scallops exposed to toxic dinoflagellates, Toxins. 13 (1) (2020) 21, https://doi.org/10.3390/toxins13010021

[57] J. M. Grim, K. A. Hyndman, T. Kriska, A. W. Girotti, E. L. Crockett, Relationship between oxidizable fatty acid content and level of antioxidant glutathione peroxidases in marine fish, J. Exp. Biol. 214 (22) (2011) 3751-3759, https://doi.org/10.1242/jeb.058214.

[58] S. Cao, Y. Wang, M. Xu, Y. Jiang, X. Wang, Y. Yang, Z. Zhou, Y. Wang, Autophagy, Apoptosis, and Ferroptosis Play Crucial Roles in Hemocyte Adaptive Mechanisms of the Marine Mussel *Mytilus Edulis* Exposed to Bde-47, Available at SSRN 5051350. (2024), https://ssrn.com/abstract=5051350

[59] Y. Duan, Y. Xiao, R. Zhu, Y. Nan, Y. Yang, X. Huang, D. Zhang, Effects of Ammonia Stress on the Antioxidant, Ferroptosis, and Immune Response in the Liver of Golden Po*mpano Trachinotus ovatus,* Antioxidants. 14 (4) (2025) 419.

[60] B. Gan, ACSL4, PUFA, and ferroptosis: new arsenal in anti-tumor immunity. *STTT*, *7* (1) (2022) 128, https://www.nature.com/articles/s41392-022-01004-z.

[61] Y. Zhou, Z. He, Q. Xu, S. Xie, P. Li, Q. Wang, J. Miao, L.Pan, Detoxification metabolic pathways and hepatotoxicity mechanisms of B [a] P in reproductive clam *Ruditapes philippinarum*, Comp. Biochem. Physiol. Part - C: Toxicol. Pharmacol.

(2025) 110378, https://doi.org/10.1016/j.cbpc.2025.110378.

[62] Q. Qiao, W. Liu, K. Wu, T. Song, J. Hu, X. Huang, J. Wen, L. Chen, X. Zhang, Female zebrafish (*Danio rerio*) are more vulnerable than males to microcystin-LR exposure, without exhibiting estrogenic effects, Aquat. Toxicol. 142 (2013) 272-282.

[63] S. Le Manach, N. Khenfech, H. Huet, Q. Qiao, C. Duval, A. Marie, G. Bolbach, G. Clodic, C. Djediat, C. Bernard, M. Edery, B. Marie, Gender-specific toxicological effects of chronic exposure to pure microcystin-LR or complex *Microcystis aeruginosa* extracts on adult medaka fish, Environ. Sci. Technol. *50*(15) (2016) 8324-8334.

[64] A. d. S. Ferrão-Filho, B. Kozlowsky-Suzuki, Cyanotoxins: bioaccumulation and effects on aquatic animals, Mar. Drugs. 9 (12) (2011) 2729-2772, https://doi.org/10.3390/md9122729.

[65] F. Jüttner, H. Lüthi, Topology and enhanced toxicity of bound microcystins in Microcystis PCC 7806, Toxicon, 51 (3) (2008) 388-397, https://doi.org/10.1016/j.toxicon.2007.10.013.

[66] R. Lazzara, D. Fernandes, M. Faria, J. F. López, R. Tauler, C. Porte, Changes in lipid content and fatty acid composition along the reproductive cycle of the freshwater mussel *Dreissena polymorpha*: its modulation by clofibrate exposure, Sci. Total Environ. 432 (2012) 195-201, https://doi.org/10.1016/j.scitotenv.2012.05.094.

[67] E. Kladchenko, A. Tkachuk, M. Podolskaya, A. Y. Andreyeva, ROS production and mitochondrial membrane potential in hemocytes of marine bivalves, *Mytilus galloprovincialis* and *Magallana gigas*, under hypoosmotic stress, Comp. Biochem. Physiol. B. 269 (2024) 110901, https://doi.org/10.1016/j.cbpb.2023.110901.

[68] J. R. Christie, T. M. Schaerf, M. Beekman, Selection against heteroplasmy explains the evolution of uniparental inheritance of mitochondria, PLOS Genet. 11 (4) (2015) e1005112, https://doi.org/10.1371/journal.pgen.1005112.

Declaration of Interest Statement

The authors declare no competing financial interest.

Graphical abstract

Highlights

- MC-LR induced mitochondrial dysfunction and oxidative stress collapse in mussel gonads.
- Ferroptosis was confirmed as the primary pathway mediating MC-LR reproductive toxicity.
- Sex-specific susceptibility was driven by the mitochondrial system under doubly uniparental inheritance.